In [16]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import BaggingClassifier,AdaBoostClassifier
from sklearn.metrics import accuracy_score,precision_score,recall_score,f1_score
```

In [17]:
```
df=pd.read_csv('/home/backup/ml_datasets/titanic.csv')
print(df.head())
print(df.isnull().sum())
```

```
   pclass  survived                                             name     sex  \
0     1.0         1                    Allen, Miss. Elisabeth Walton  female
1     1.0         1                   Allison, Master. Hudson Trevor    male
2     1.0         0                     Allison, Miss. Helen Loraine  female
3     1.0         0             Allison, Mr. Hudson Joshua Creighton    male
4     1.0         0  Allison, Mrs. Hudson J C (Bessie Waldo Daniels)  female

       age  sibsp  parch  ticket      fare    cabin embarked boat   body  \
0  29.0000    0.0    0.0   24160  211.3375       B5        S    2    NaN
1   0.9167    1.0    2.0  113781  151.5500  C22 C26        S   11    NaN
2   2.0000    1.0    2.0  113781  151.5500  C22 C26        S  NaN    NaN
3  30.0000    1.0    2.0  113781  151.5500  C22 C26        S  NaN  135.0
4  25.0000    1.0    2.0  113781  151.5500  C22 C26        S  NaN    NaN

                        home.dest
0                     St Louis, MO
1  Montreal, PQ / Chesterville, ON
2  Montreal, PQ / Chesterville, ON
3  Montreal, PQ / Chesterville, ON
4  Montreal, PQ / Chesterville, ON
pclass          0
survived        0
name            0
sex             0
age           263
sibsp           0
parch           0
ticket          0
fare            1
cabin        1014
embarked        2
boat          823
body         1188
home.dest     564
dtype: int64
```

In [18]:
```
df=df[['survived','pclass','sex','age','sibsp','parch','fare','embarked']]
df['age']=df['age'].fillna(df['age'].median())
df['fare']=df['fare'].fillna(df['fare'].median())
df['embarked']=df['embarked'].fillna(df['embarked'].mode()[0])

df['sex']=df['sex'].map({'male':0,'female':1})
df=pd.get_dummies(df,columns=['embarked'],dtype=int)

X=df.drop('survived',axis=1)
y=df['survived']

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
```

In [20]:
```
tree=DecisionTreeClassifier(max_depth=2,random_state=42)
bagging=BaggingClassifier(base_estimator=tree,n_estimators=50,random_state=42)
bagging.fit(X_train,y_train)
```

```
y_pred_bagging=bagging.predict(X_test)

print("----Bagging-----")
print("Accuracy:",accuracy_score(y_test,y_pred_bagging))
print("Precision:",precision_score(y_test,y_pred_bagging))
print("Recall:",recall_score(y_test,y_pred_bagging))
print("F1:",f1_score(y_test,y_pred_bagging))
```

```
----Bagging-----
Accuracy: 0.7290076335877863
Precision: 0.873015873015873
Recall: 0.4661016949152542
F1: 0.6077348066298341
```

In [22]:

```
boosting=AdaBoostClassifier(base_estimator=DecisionTreeClassifier(max_depth=1),
                            n_estimators=50,random_state=42)
boosting.fit(X_train,y_train)
y_pred_boosting=boosting.predict(X_test)
print("----Boosting-----")
print("Accuracy:",accuracy_score(y_test,y_pred_boosting))
print("Precision:",precision_score(y_test,y_pred_boosting))
print("Recall:",recall_score(y_test,y_pred_boosting))
print("F1:",f1_score(y_test,y_pred_boosting))
```

```
----Boosting-----
Accuracy: 0.7862595419847328
Precision: 0.8229166666666666
Recall: 0.6694915254237288
F1: 0.7383177570093459
```

In [ ]: